Emergency Department Staffing Optimization

Problem Overview

We aim to optimize staffing levels in an Emergency Department (ED) to ensure timely patient care while balancing cost and resource utilization. The ED is modeled as a queuing system with stochastic arrivals and service times.

System Components

Patient Arrival Process

Arrivals follow a non-homogeneous Poisson process with rate $\lambda(t)$ that varies by time of day (morning, afternoon, night).

Triage Levels

Each patient is assigned a triage category with a fixed probability:

$$P(\text{Level } i) = p_i.$$

We consider two levels (Urgent, Non-urgent) :

$$p_2 = 0.40, \quad p_3 = 0.60.$$

Service Times

Service time for a triage level i is assumed exponential with rate μ_i:

$$S_i \sim \text{Exp}(\mu_i), \qquad \mathbb{E}[S_i] = \frac{1}{\mu_i}.$$

Example service rates (Critical, Urgent, Non-urgent):

$$\mu_2 = 4,\ \mu_3 = 6.$$

Higher acuity = shorter mean service time but stricter wait limits.

Staffing Policies

- **Static Staffing:** Fixed number of staff for the whole horizon.
- **Dynamic Staffing:** Staff levels change by shift to match $\lambda(t)$.
- **Threshold-Based:** Add or remove staff when queue length or waiting time exceeds a threshold.

Performance Metrics

Average Waiting Time

$$\bar{W} = \frac{1}{N}\sum_{j=1}^{N} W_j,$$

with violations tracked when $W_j > W^{\max}_{\mathrm{triage}(j)}$.

Staffing Cost

If $N(t)$ staff are scheduled with cost rate c_s:

$$C = c_s \int_0^T N(t)\, dt.$$

Utility Function

A combined objective:

$$U = -w_1\bar{W} - w_3 C.$$

Simulation and Optimization

A discrete-event simulation evaluates each policy by reproducing arrivals, queuing, and service. We compare policies via utility U and optionally apply:

- Static approach : Apply the same number of staff for every state.
- Greedy approach: Apply the number of staff based on the number of patients left in the queue.
- Reinforcement Learning approach: Apply MDP algorithm for the problem, using simulation to get rewards.

The goal is to identify staffing rules that maximize U and keep waiting times within triage limits.

Initial simulation environment

For training, computing and comparing every approaches, we use the following parameters:

$$\begin{aligned}
\text{Queue Capacity: } M &= 20 \\
\text{Maximum Staff: } S &= 10 \\
\text{Arrival rate: } \mu &= 10 \\
\text{Service rate: } \lambda_2 &= 4.0 \\
\lambda_3 &= 6.0 \\
\text{Reward for urgent } r_2 &= 50 \\
\text{Reward for non-urgent } r_3 &= 20 \\
\text{Reward for Maintenance } c_m &= 10 \\
\text{Reward for ca } c_a &= 50 \\
\text{Penalty for blocking } c_p &= 100
\end{aligned}$$

This is just the initial parameter. We will change them later for analyze specific properties of each approaches.

Static Approach

We keep the same number of staffs for every states of the problem.

Optimal solution

By running simulation for different number of staffs, the optimal number of staffs for the above environment is 3. This also align with the theoretical optimal value.

Theoretical Optimal

Important properties

Given n i.d.p random variable $X_1, X_2, \ldots, X_n$ where $X_i \sim Exp(\lambda_i)$, we have:

$$\min(X_1, \ldots, X_n) \sim Exp(\sum_{i=1}^{n} \lambda_i) \quad \text{(Proven in class)}$$

Therefore, in a system consist of n busy servers with service rate of λ, the transition rate to completeing 1 task is $n\lambda$.

General Double-Class system

Let the number of staff in this policy is s. We define state space as (w_2, w_3, s_2, s_3) where w_2, w_3, s_2, s_3 are the number of urgent, non-urgent patients in queue and number of urgent, non-urgent patients in service. Then, we will have the following transition:

$$
\begin{aligned}
(w_2, w_3, s_2, s_3) &\xrightarrow{p_2\mu} (w_2+1, w_3, s_2, s_3) && (s_2+s_3=s, w_2+w_3<M)\\
(w_2, w_3, s_2, s_3) &\xrightarrow{p_3\mu} (w_2, w_3+1, s_2, s_3) && (s_2+s_3=s, w_2+w_3<M)\\
(w_2, w_3, s_2, s_3) &\xrightarrow{s_2\lambda_2} (w_2-1, w_3, s_2, s_3) && (s_2+s_3=s, w_2>0)\\
(w_2, w_3, s_2, s_3) &\xrightarrow{s_3\lambda_3} (w_2-1, w_3, s_2+1, s_3-1) && (s_2+s_3=s, w_2>0, s_3>0)\\
(0, w_3, s_2, s_3) &\xrightarrow{s_2\lambda_2} (0, w_3-1,\ s_2+1, s_3-1) && (s_2+s_3=s, w_3>0)\\
(0, w_3, s_2, s_3) &\xrightarrow{s_3\lambda_3} (0, w_3-1, s_2, s_3) && (s_2+s_3=s, w_3>0, s_3>0)\\
(0, 0, s_2, s_3) &\xrightarrow{p_2\mu} (0, 0, s_2+1, s_3) && (s_2+s_3<s)\\
(0, 0, s_2, s_3) &\xrightarrow{p_3\mu} (0, 0, s_2, s_3+1) && (s_2+s_3<s)\\
(0, 0, s_2, s_3) &\xrightarrow{s_2\lambda_2} (0, 0, s_2-1, s_3) && (s_2>0)\\
(0, 0, s_2, s_3) &\xrightarrow{s_3\lambda_3} (0, 0, s_2, s_3-1) && (s_3>0)\\
(w_2, w_3, s_2, s_3) &\xrightarrow{\mu} (w_2, w_3, s_2, s_3) && (w_2+w_3=M, s_2+s_3=s)
\end{aligned}
$$

Let $P(w_2, w_3, s_2, s_3)$ denote the state probability. Thus, balance equations:

Interior states $(s_2+s_3=s,\ w_2>0,\ w_2+w_3<M)$:

$$
\begin{aligned}
P(w_2, w_3, s_2, s_3)\big(p_2\mu+p_3\mu+s_2\lambda_2+s_3\lambda_3\big) = {} & P(w_2-1, w_3, s_2, s_3)\, p_2\mu \\
& + P(w_2, w_3-1, s_2, s_3)\, p_3\mu \\
& + P(w_2+1, w_3, s_2, s_3)\, s_2\lambda_2 \\
& + P(w_2+1, w_3, s_2-1, s_3+1)\, s_3\lambda_3
\end{aligned}
$$

Full queue state $(s_2+s_3=s,\ w_2+w_3=M,\ w_2>0)$:

$$
\begin{aligned}
P(w_2, w_3, s_2, s_3)\big(s_2\lambda_2+s_3\lambda_3\big) = {} & P(w_2+1, w_3, s_2, s_3)\, s_2\lambda_2 \\
& + P(w_2+1, w_3, s_2-1, s_3+1)\, s_3\lambda_3
\end{aligned}
$$

Priority boundary state $(s_2+s_3=s,\ w_2=0,\ w_3>0)$:

$$
\begin{aligned}
P(0, w_3, s_2, s_3)\big(p_3\mu+s_2\lambda_2+s_3\lambda_3\big) = {} & P(0, w_3-1, s_2, s_3)\, p_3\mu \\
& + P(1, w_3, s_2, s_3)\, s_2\lambda_2 \\
& + P(1, w_3, s_2-1, s_3+1)\, s_3\lambda_3
\end{aligned}
$$

Empty queue, servers not full ($w_2 = 0,\ w_3 = 0,\ 0 < s_2 + s_3 < s$):

$$\begin{aligned} P(0,0,s_2,s_3)\big(p_2\mu + p_3\mu + s_2\lambda_2 + s_3\lambda_3\big) &= P(0,0,s_2-1,s_3)\,p_2\mu \\ &\quad + P(0,0,s_2,s_3-1)\,p_3\mu \\ &\quad + P(0,0,s_2+1,s_3)\,s_2\lambda_2 \\ &\quad + P(0,0,s_2,s_3+1)\,s_3\lambda_3 \end{aligned}$$

Empty queue, servers full ($w_2 = w_3 = 0,\ s_2 + s_3 = s$):

$$\begin{aligned} P(0,0,s_2,s_3)\big(s_2\lambda_2 + s_3\lambda_3\big) &= P(0,0,s_2+1,s_3)\,s_2\lambda_2 \\ &\quad + P(0,0,s_2,s_3+1)\,s_3\lambda_3 \end{aligned}$$

Since the system is too big and might be unsolvable, I will approximate the system of double-class system to a single-class system for easier proof. We can approximate the parameters of this system since the arrival rate is moderate compared to the queue capacity and service rate.

Reduced single-class system

The expected service time of both urgent and non-urgent patients is:

$$\mathbb{E}[S_t] \approx p_2\frac{1}{\lambda_2} + p_3\frac{1}{\lambda_3} = 0.4 \cdot \frac{1}{4.0} + 0.6 \cdot \frac{1}{6.0} = 0.1 + 0.1 = 0.2.$$

Therefore, the approximate service rate is:

$$\lambda = \frac{1}{\mathbb{E}[S_t]} \approx 5.$$

The expected reward of both urgent and non-urgent time:

$$E[\text{reward}] = 50 * 0.4 + 20 * 0.6 = 32$$

Therefore, we reduce the problem to the following single-class:

$$S = s, \qquad M = 20, \qquad K = S + M = 20 + s,$$

$$\text{arrival rate } \mu = 10, \qquad \text{service rate } \lambda = 5.$$

Similarly, we define the state space as w, s where w is number of patients in queue and s is number of patients in service. Denote P(w, s) the stationary distribution of those state. We can see that there are no states such that $w > 0$ and $s < \max s$. Therefore, instead of represent (w, s), we can denoted them using a single variable n where $n = w + s$. Therefore, we have the condition:

$$0 \le n \le K$$

Then, we obtain the balance equations:

Interior states $(1 \le n \le K-1)$:

$$\mu P(n-1) = \min(n,s)\lambda P(n).$$

Boundary states:

$$\mu P(0) = \lambda P(1),$$
$$\mu P(K-1) = s\,\lambda P(K).$$

We have the relation:

$$P(n) = \frac{\mu}{\min(n,s)\,\mu_{\text{staff}}}\,P(n-1).$$

We solve this recursion by considering two cases.

If $0 \le n \le s$

Since $\min(n,s) = n$, and therefore:

$$P(n) = \frac{\mu}{n\,\mu_{\text{staff}}} P(n-1)$$

$$\Rightarrow P(n) = \frac{\mu^n}{n!\,\mu_{\text{staff}}^n}\,P(0), \qquad 0 \le n \le s.$$

If $s < n \le K$

Since $\min(n,s) = s$, and therefore:

$$P(n) = \frac{\mu}{s\,\mu_{\text{staff}}} P(n-1)$$

With Case 1, we also have:

$$P(n) = \frac{\mu^n}{s!\,s^{\,n-s}\,\mu_{\text{staff}}^n}\,P(0), \qquad s < n \le K$$

Combine this, we have:

$$P(n) = \begin{cases} \dfrac{\mu^n}{n!\,\mu_{\text{staff}}^n}\,P(0), & 0 \le n \le s \\ \dfrac{\mu^n}{s!\,s^{\,n-s}\,\mu_{\text{staff}}^n}\,P(0), & s < n \le K \end{cases}$$

Normalize the sum to 1, we have:

$$\frac{1}{P(0)} = \sum_{n=0}^{s} \frac{\mu^n}{n!\,\mu_{\text{staff}}^n} + \sum_{n=s+1}^{K} \frac{\mu^n}{s!\,s^{\,n-s}\,\mu_{\text{staff}}^n}.$$

The blocking probability is:

$$P_{\text{block}} = P(K) = \frac{\mu^K}{s!\,s^{\,K-s}\,\mu_{\text{staff}}^K}\,P(0)$$

Then the rate of entering system is:

$$\mu_{\text{eff}} = \mu\big(1 - P_{\text{block}}\big)$$

The expected reward with staff level s is:

$$\begin{aligned} R(s) &= E[\text{reward}] \cdot \mu_{\text{eff}} - c_m \cdot s - c_p \cdot \mu P_{\text{block}} \\ &= 32 \cdot \mu\big(1 - P_{\text{block}}(s)\big) - 10s - 100 \cdot \mu P_{\text{block}}(s). \end{aligned}$$

Consider s from 1 to 10, we have:

$$\arg\max_{1 \le s \le 10} R(s) = 3$$

Greedy Approach

The idea of this approach is to dynamically change the number of staff accordingly to the number of patients in the queue. The policy for this is as follow. Let P_q the number of patient in queue.

- Increase when the queue is likely to be overloaded ($P_q \geq v_o$)
- Decrease when the queue is free ($P_q \leq v_f$)

The threshold will be chosen to:

- Minimizing the cost of maintenance (v_f)
- Minimizing the penalty of blocking (v_o)

Also, by only increase and decrease the number of staffs when needed, it will also minimize the cost of increasing a staff.

Reinforcement Learning Approach